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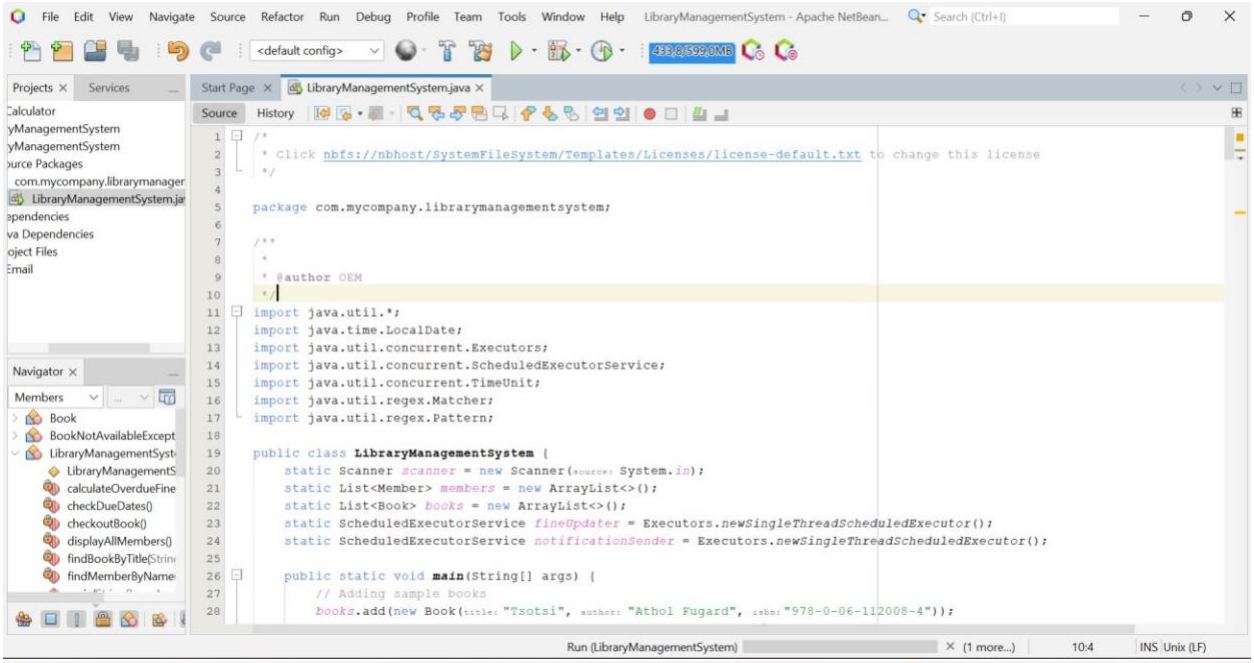
20231494

JD521FA2-3

# Contents

Screenshot of code ..... 3
OUTPUT ..... 11
DECLARATION OF AUTHENTICITY ..... 18

# Screenshot of code



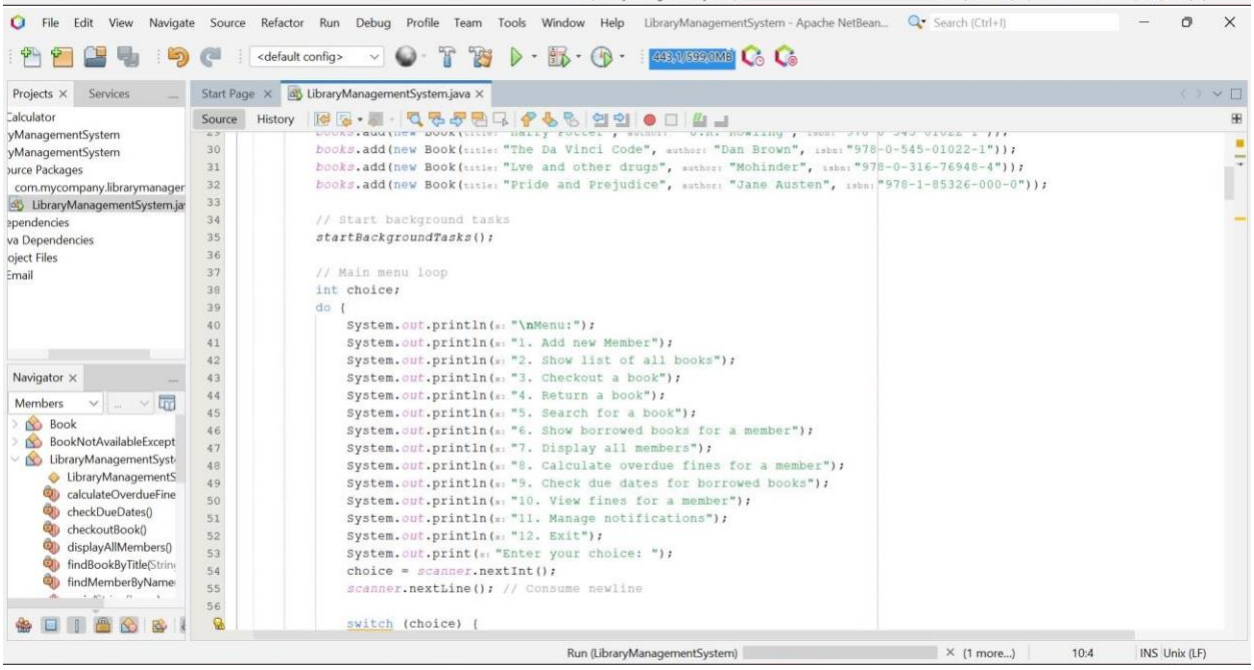
```java
/*
 * Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
 */

package com.mycompany.librarymanagementsystem;

/**
 *
 * @author OEM
 */
import java.util.*;
import java.time.LocalDate;
import java.util.concurrent.Executors;
import java.util.concurrent.ScheduledExecutorService;
import java.util.concurrent.TimeUnit;
import java.util.regex.Matcher;
import java.util.regex.Pattern;

public class LibraryManagementSystem {
    static Scanner scanner = new Scanner(source: System.in);
    static List<Member> members = new ArrayList<>();
    static List<Book> books = new ArrayList<>();
    static ScheduledExecutorService fineUpdater = Executors.newSingleThreadScheduledExecutor();
    static ScheduledExecutorService notificationSender = Executors.newSingleThreadScheduledExecutor();

    public static void main(String[] args) {
        // Adding sample books
        books.add(new Book(title: "Tsotsi", author: "Athol Fugard", isbn: "978-0-06-112008-4"));
```



```java
        books.add(new Book(title: "The Da Vinci Code", author: "Dan Brown", isbn: "978-0-545-01022-1"));
        books.add(new Book(title: "Lve and other drugs", author: "Mohinder", isbn: "978-0-316-76948-4"));
        books.add(new Book(title: "Pride and Prejudice", author: "Jane Austen", isbn: "978-1-85326-000-0"));

        // Start background tasks
        startBackgroundTasks();

        // Main menu loop
        int choice;
        do {
            System.out.println(x: "\nMenu:");
            System.out.println(x: "1. Add new Member");
            System.out.println(x: "2. Show list of all books");
            System.out.println(x: "3. Checkout a book");
            System.out.println(x: "4. Return a book");
            System.out.println(x: "5. Search for a book");
            System.out.println(x: "6. Show borrowed books for a member");
            System.out.println(x: "7. Display all members");
            System.out.println(x: "8. Calculate overdue fines for a member");
            System.out.println(x: "9. Check due dates for borrowed books");
            System.out.println(x: "10. View fines for a member");
            System.out.println(x: "11. Manage notifications");
            System.out.println(x: "12. Exit");
            System.out.print(x: "Enter your choice: ");
            choice = scanner.nextInt();
            scanner.nextLine(); // Consume newline

            switch (choice) {
```

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help LibraryManagementSystem - Apache NetBean...

Projects × Services

Start Page × LibraryManagementSystem.java ×

Source History

```java
                case 1:
                    registerMember();
                    break;
                case 2:
                    showAllBooks();
                    break;
                case 3:
                    checkoutBook();
                    break;
                case 4:
                    returnBook();
                    break;
                case 5:
                    searchBook();
                    break;
                case 6:
                    showBorrowedBooks();
                    break;
                case 7:
                    displayAllMembers();
                    break;
                case 8:
                    calculateOverdueFines();
                    break;
                case 9:
                    checkDueDates();
                    break;
                case 10:
```

Navigator × Members

Snipping Tool

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Run (LibraryManagementSystem)

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Start Page × LibraryManagementSystem.java ×

Source History

```java
                    viewFines();
                    break;
                case 11:
                    manageNotifications();
                    break;
                case 12:
                    stopBackgroundTasks(); // Stop background tasks before exiting
                    System.out.println("Exiting the program...");
                    break;
                default:
                    System.out.println("Invalid choice. Please try again.");
            }
        } while (choice != 12);
    }

    private static void startBackgroundTasks() {
        fineUpdater.scheduleAtFixedRate(() -> {
            System.out.println("\nUpdating fines...");
            updateFines();
        }, initialDelay: 0, period: 1, unit: TimeUnit.DAYS); // Update fines every day

        notificationSender.scheduleAtFixedRate(() -> {
            System.out.println("\nSending notifications...");
            sendNotifications();
        }, initialDelay: 0, period: 7, unit: TimeUnit.DAYS); // Send notifications every
    }

    private static void stopBackgroundTasks() {
```

Snipping Tool

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Run (LibraryManagementSystem)

```java
        fineUpdater.shutdown();
        notificationSender.shutdown();
    }

    private static void registerMember() {
        System.out.print("Enter member name: ");
        String name = scanner.nextLine();
        System.out.print("Enter member email: ");
        String email = scanner.nextLine();
        Member member = new Member(name, email);
        members.add(member);
        System.out.println("Member added successfully.");
    }

    private static void showAllBooks() {
        System.out.println("\nList of all books:");
        for (Book book : books) {
            System.out.println(book.getTitle() + " by " + book.getAuthor() + " (
        }
    }

    private static void checkoutBook() {
        System.out.print("Enter member name: ");
        String memberName = scanner.nextLine();
        Member member = findMemberByName(name: memberName);
        if (member == null) {
            System.out.println("Member not found.");
            return;
        }
```

```java
        }

        System.out.print("Enter title of the book to checkout: ");
        String bookTitle = scanner.nextLine();
        Book bookToCheckout = findBookByTitle(title: bookTitle);
        if (bookToCheckout == null) {
            System.out.println("Book not found.");
            return;
        }

        try {
            member.borrowBook(book: bookToCheckout);
            System.out.println("Book '" + bookToCheckout.getTitle() + "' checked out successfully.");
        } catch (BookNotAvailableException e) {
            System.out.println("Error: " + e.getMessage());
        }
    }

    private static void returnBook() {
        System.out.print("Enter member name: ");
        String memberName = scanner.nextLine();
        Member member = findMemberByName(name: memberName);
        if (member == null) {
            System.out.println("Member not found.");
            return;
        }

        System.out.print("Enter title of the book to return: ");
```

```java
        String bookTitle = scanner.nextLine();
        System.out.print(s: "Enter ISBN of the book to return: ");
        String isbn = scanner.nextLine();

        member.returnBook(title: bookTitle, isbn);
    }

    private static void searchBook() {
        System.out.print(s: "Enter search query (title, author, or ISBN): ");
        String query = scanner.nextLine().toLowerCase();

        System.out.println(x: "\nSearch results:");
        for (Book book : books) {
            if (book.getTitle().toLowerCase().contains(s: query)
                    || book.getAuthor().toLowerCase().contains(s: query)
                    || book.getIsbn().toLowerCase().contains(s: query)) {
                System.out.println(book.getTitle() + " by " + book.getAuthor() +
            }
        }
    }

    private static void showBorrowedBooks() {
        System.out.print(s: "Enter member name: ");
        String memberName = scanner.nextLine();
        Member member = findMemberByName(name: memberName);
        if (member == null) {
            System.out.println(x: "Member not found.");
            return;
```

Snipping Tool

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```java
        }

        List<Book> borrowedBooks = member.getBorrowedBooks();
        if (borrowedBooks.isEmpty()) {
            System.out.println("No books borrowed by " + memberName);
        } else {
            System.out.println("Borrowed books by " + memberName + ":");
            for (Book book : borrowedBooks) {
                System.out.println(book.getTitle() + " by " + book.getAuthor() + " (ISBN: " + book.getIsbn() + ")");
            }
        }
    }

    private static void displayAllMembers() {
        System.out.println(x: "\nList of all members:");
        for (Member member : members) {
            System.out.println(member.getName() + " (" + member.getEmail() + ")");
        }
    }

    private static void calculateOverdueFines() {
        System.out.print(s: "Enter member name: ");
        String memberName = scanner.nextLine();
        Member member = findMemberByName(name: memberName);
        if (member == null) {
            System.out.println(x: "Member not found.");
            return;
        }
```

File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help LibraryManagementSystem - Apache NetBean...

Start Page | LibraryManagementSystem.java

```java
        }

        double totalFines = member.calculateTotalOverdueFines();
        System.out.println("Total overdue fines for " + memberName + ": $" + totalFines);
    }

    private static void checkDueDates() {
        System.out.println("\nChecking due dates for borrowed books...");
        for (Member member : members) {
            List<Book> borrowedBooks = member.getBorrowedBooks();
            for (Book book : borrowedBooks) {
                System.out.println("Book '" + book.getTitle() + "' is due on " + book.getDueDate());
            }
        }
    }

    private static void viewFines() {
        System.out.print("Enter member name: ");
        String memberName = scanner.nextLine();
        Member member = findMemberByName(memberName);
        if (member == null) {
            System.out.println("Member not found.");
            return;
        }

        double totalFines = member.calculateTotalOverdueFines();
        System.out.println("Total overdue fines for " + memberName + ": $" + totalFines);
    }
```

Run (LibraryManagementSystem) | 123:24 | INS Unix (LF)

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Start Page | LibraryManagementSystem.java

```java
    }

    private static void manageNotifications() {
        System.out.println("\nManaging notifications...");
        // Logic for managing notifications can be added here
    }

    private static Member findMemberByName(String name) {
        for (Member member : members) {
            if (member.getName().equalsIgnoreCase(name)) {
                return member;
            }
        }
        return null;
    }

    private static Book findBookByTitle(String title) {
        for (Book book : books) {
            if (book.getTitle().equalsIgnoreCase(title)) {
                return book;
            }
        }
        return null;
    }

    private static void updateFines() {
        System.out.println("Fines updated.");
    }
```

Snipping Tool

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Run (LibraryManagementSystem) | 123:24 | INS Unix (LF)

```java

    private static void sendNotifications() {
        System.out.println("Notifications sent.");
    }
}

class Book {
    String title;
    String author;
    String isbn;
    private boolean isAvailable;
    private LocalDate dueDate;

    public Book(String title, String author, String isbn) {
        this.title = title;
        this.author = author;
        this.isbn = isbn;
        this.isAvailable = true;
        this.dueDate = null;
    }

    public String getTitle() {
        return title;
    }

    public String getAuthor() {
        return author;
    }
```

```java
    }

    public boolean isAvailable() {
        return isAvailable;
    }

    public void toggleAvailability() {
        isAvailable = !isAvailable;
    }

    public LocalDate getDueDate() {
        return dueDate;
    }

    public void setDueDate(LocalDate dueDate) {
        this.dueDate = dueDate;
    }

    public double calculateOverdueFine() {
        if (dueDate != null && LocalDate.now().isAfter(other: dueDate)) {
            long daysOverdue = LocalDate.now().toEpochDay() - dueDate.toEpochDay();
            double fine = daysOverdue * 0.50; // Example: $0.50 per day overdue
            return fine;
        }
        return 0;
    }
}

class Member {
```

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Start Page | LibraryManagementSystem.java

```java
    String name;
    private String email;
    List<Book> borrowedBooks;

    public Member(String name, String email) {
        this.name = name;
        if (isValidEmail(email)) {
            this.email = email;
        } else {
            throw new IllegalArgumentException("Invalid email format.");
        }
        this.borrowedBooks = new ArrayList<>();
    }

    private boolean isValidEmail(String email) {
        String regex = "^(.+)@(.+)$";
        Pattern pattern = Pattern.compile(regex);
        Matcher matcher = pattern.matcher(email);
        return matcher.matches();
    }

    public String getName() {
        return name;
    }

    public String getEmail() {
        return email;
    }
```

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Start Page | LibraryManagementSystem.java

```java
    }

    public List<Book> getBorrowedBooks() {
        return borrowedBooks;
    }

    public void borrowBook(Book book) throws BookNotAvailableException {
        if (!book.isAvailable()) {
            throw new BookNotAvailableException("The book is not available for checkout.");
        }
        borrowedBooks.add(book);
        book.toggleAvailability();
        book.setDueDate(LocalDate.now().plusDays(14)); // Example loan period: 14 days
    }

    public void returnBook(String title, String isbn) {
        Book bookToReturn = null;
        for (Book book : borrowedBooks) {
            if (book.getTitle().equalsIgnoreCase(title) && book.getIs
                bookToReturn = book;
                break;
            }
        }
        if (bookToReturn != null) {
            borrowedBooks.remove(bookToReturn);
            bookToReturn.toggleAvailability();
            System.out.println("Book returned successfully.");
        } else {
```

Snipping Tool

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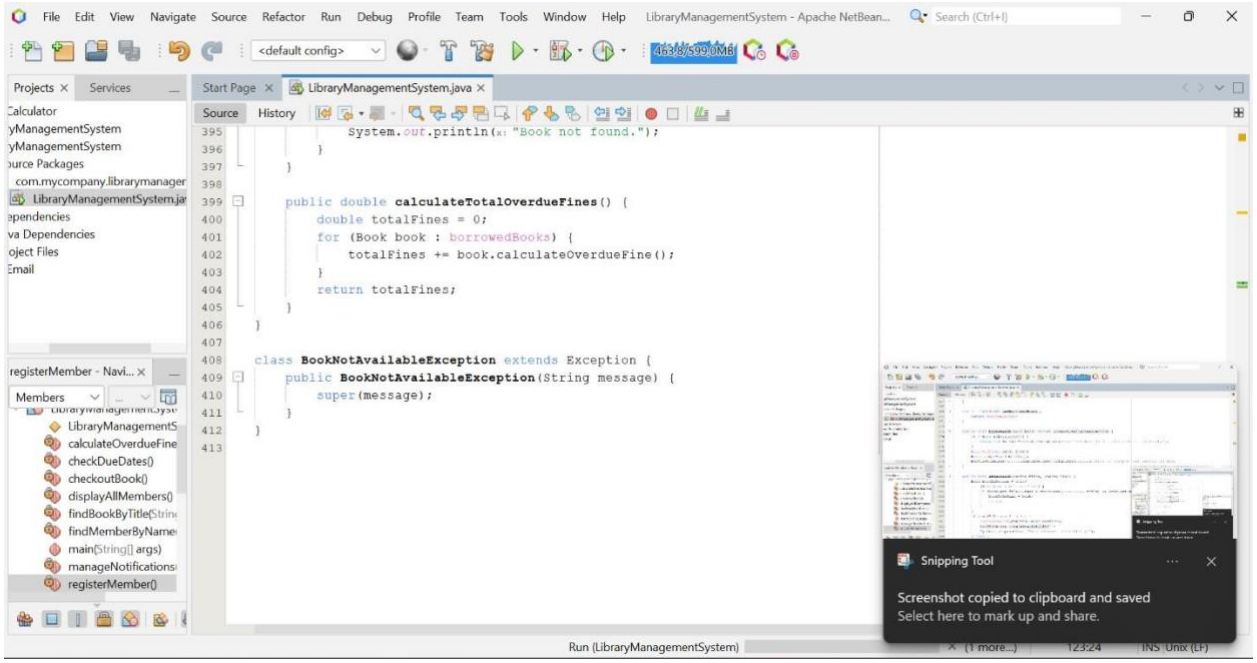
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<default config>
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Projects
Services
Start Page
LibraryManagementSystem.java
Source
History
System.out.println("Book not found.");
}
}
public double calculateTotalOverdueFines() {
double totalFines = 0;
for (Book book : borrowedBooks) {
totalFines += book.calculateOverdueFine();
}
return totalFines;
}
}
class BookNotAvailableException extends Exception {
public BookNotAvailableException(String message) {
super(message);
}
}
registerMember - Navi...
Members
calculateOverdueFine
checkDueDates()
checkoutBook()
displayAllMembers()
findBookByTitle(Strin
findMemberByName
main(String[] args)
manageNotifications
registerMember()
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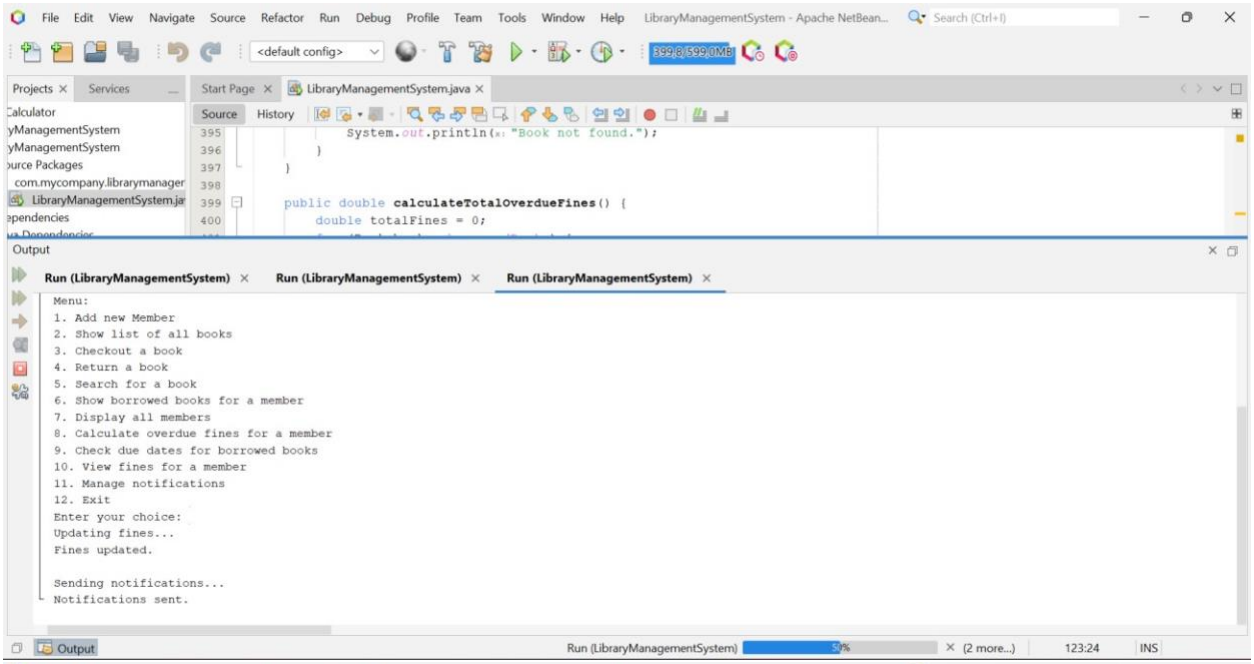
Run (LibraryManagementSystem)
123:24
INS Unix (LF)

# OUTPUT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ource Packages
com.mycompany.librarymanager
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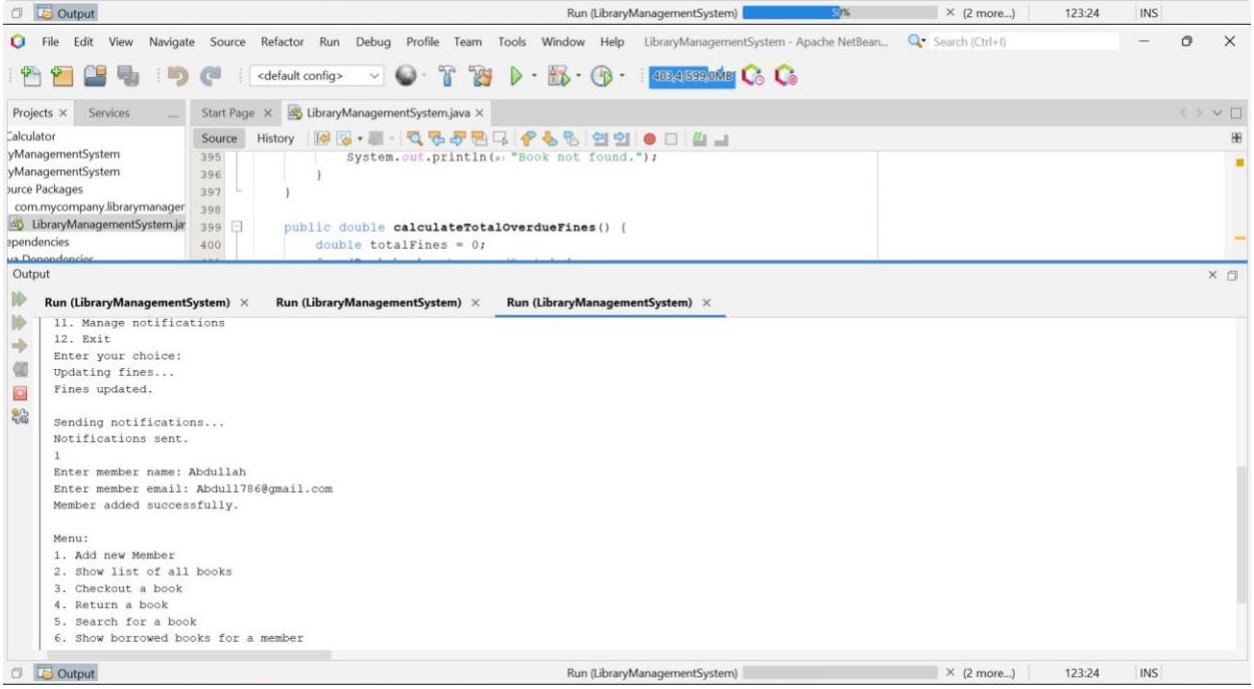
ependencies
System.out.println("Book not found.");
}
}
public double calculateTotalOverdueFines() {
double totalFines = 0;
Output
Run (LibraryManagementSystem)
Run (LibraryManagementSystem)
Run (LibraryManagementSystem)
Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice:
Updating fines...
Fines updated.
Sending notifications...
Notifications sent.
Output
Run (LibraryManagementSystem)
(2 more...)
123:24
INS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ource Packages
com.mycompany.librarymanager
LibraryManagementSystem.jar
ependencies
System.out.println("Book not found.");
}
}
public double calculateTotalOverdueFines() {
double totalFines = 0;
Output
Run (LibraryManagementSystem)
Run (LibraryManagementSystem)
Run (LibraryManagementSystem)
11. Manage notifications
12. Exit
Enter your choice:
Updating fines...
Fines updated.
Sending notifications...
Notifications sent.
1
Enter member name: Abdullah
Enter member email: Abdull786@gmail.com
Member added successfully.
Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
Output
Run (LibraryManagementSystem)
(2 more...)
123:24
INS

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
11. Manage notifications
12. Exit
Enter your choice: 2

List of all books:
Tsotsi by Athol Fugard (ISBN: 978-0-06-112008-4)
Harry Potter by  J.K. Rowling (ISBN: 978-0-545-01022-1)
The Da Vinci Code by Dan Brown (ISBN: 978-0-545-01022-1)
Lve and other drugs by Mohinder (ISBN: 978-0-316-76948-4)
Pride and Prejudice by Jane Austen (ISBN: 978-1-85326-000-0)

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Harry Potter by  J.K. Rowling (ISBN: 978-0-545-01022-1)
The Da Vinci Code by Dan Brown (ISBN: 978-0-545-01022-1)
Lve and other drugs by Mohinder (ISBN: 978-0-316-76948-4)
Pride and Prejudice by Jane Austen (ISBN: 978-1-85326-000-0)

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 3
Enter member name:
```

```
395            System.out.println("Book not found.");
396        }
397    }
398
399    public double calculateTotalOverdueFines() {
400        double totalFines = 0;
```

```
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 3
Enter member name: abdullah
Enter title of the book to checkout: tsotsi
Book 'Tsotsi' checked out successfully.

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
```

```
395            System.out.println("Book not found.");
396        }
397    }
398
399    public double calculateTotalOverdueFines() {
400        double totalFines = 0;
```

```
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 4
Enter member name: abdullah
Enter title of the book to return: tsotsi
Enter ISBN of the book to return: 978-0-06-112008-4
Book returned successfully.

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
```

```
Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 5
Enter search query (title, author, or ISBN): tsotsi

Search results:
Tsotsi by Athol Fugard (ISBN: 978-0-06-112008-4)
```

```
Tsotsi by Athol Fugard (ISBN: 978-0-06-112008-4)

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 6
Enter member name: abdullah
Borrowed books by abdullah:
Harry Potter by  J.K. Rowling (ISBN: 978-0-545-01022-1)
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 7

List of all members:
Abdullah (Abdull786@gmail.com)

Menu:
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Abdullah (Abdull786@gmail.com)

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 8
Enter member name: abdullah
Total overdue fines for abdullah: $0.0

Menu:
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit

Enter your choice: 9

Checking due dates for borrowed books...
Book 'Harry Potter' is due on 2024-05-24
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Checking due dates for borrowed books...
Book 'Harry Potter' is due on 2024-05-24

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 10
Enter member name: abdullah
Total overdue fines for abdullah: $0.0
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
Enter your choice: 10
Enter member name: abdullah
Total overdue fines for abdullah: $0.0

Menu:
1. Add new Member
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 11

Managing notifications...
```

```
            System.out.println("Book not found.");
        }
    }

    public double calculateTotalOverdueFines() {
        double totalFines = 0;
```

```
2. Show list of all books
3. Checkout a book
4. Return a book
5. Search for a book
6. Show borrowed books for a member
7. Display all members
8. Calculate overdue fines for a member
9. Check due dates for borrowed books
10. View fines for a member
11. Manage notifications
12. Exit
Enter your choice: 12
Exiting the program...
------------------------------------------------------------------------
BUILD SUCCESS
------------------------------------------------------------------------
Total time:  05:40 min
Finished at: 2024-05-10T16:20:58+02:00
------------------------------------------------------------------------
```

# DECLARATION OF AUTHENTICITY

I Chane Marais

(FULL NAME) hereby declare that the contents of this assignment are entirely my own work with the exception of the following elements: (List the elements of work in this project that were not self-generated as well as who the originator of the element is) Element Originator

Signature: CM. Date: 10-06-2024